Vallorée CA Copilot

Applied AI Engineer written challenge · a tool that drafts three on-brand WeChat messages for luxury Client Advisors · November 2026

95% of v3 options are OK to send as-is (v1: 76%)	**1/150** unacceptable in v3 (v1: 19/150)	**43/50** cases where all three options are OK (v1: 23/50)	**6 / 50** scenarios / test cases, 26 core · 13 edge · 11 adversarial

What it is. A Python tool (`ca_copilot/`) with a local web app (FastAPI + single-page frontend). Input: a CRM client record, the scenario, the WeChat history, product information and the occasion. Output: three Chinese WeChat drafts that take *different angles*, each with a rationale, any remaining compliance flags, and a timing warning when sending now would break a client preference. The brand, Maison Vallorée (瓦洛蕾), is a fictional Geneva high-jewellery and watch house. Everything in this submission is fictional.

How it got here. Three tested versions. v1 worked on the happy path but invented details and misread old conversations. v2 fixed the context problems. v3 added a numbered fact sheet and a fact-check step, which cut unacceptable drafts from 19 to 1 out of 150. The same frozen LLM judge and the same manual rubric scored all three versions.

Files: `deliverables/test_cases_v3.xlsx` (C: 50 cases with outputs; D: per-option scores; E: v1 vs v2 vs v3) · `docs/architecture.svg` · code and data in the repository · run it: `./start.sh` → http://localhost:8000.

A. Work log

The AI assistant was Claude Code (Claude Opus 5.5) in VS Code for design, coding and analysis. DeepSeek (`deepseek-chat`) is the model inside the tool. Under each step is the prompt I gave Claude Code for it.

Step	What I did	Input	Output	Tool or agent used
1	Briefed the AI as an engineering partner: goal, fixed decisions (DeepSeek via OpenAI-compatible API with the key kept in .env; fictional high-jewellery & watch brand; conda env; deliverables), and working rules (stop after each stage to report; deterministic logic in code, not the LLM; freeze each version in snapshots/; human labels drafted by AI are marked “to be checked by me”). Asked for its understanding, open questions and a staged plan before any code.	Brief PDF	Requirements summary, open questions, staged plan with outputs and acceptance criteria per stage	Claude Code (Claude Opus 5.5) in VS Code

Prompt for step 1:

```
你是我的工程搭档。我要完成一个 Applied AI Engineer 的 take-home（brief 见附件，保密，不要外传）。

目标：做一个工具，为奢侈品 Client Advisor 生成 3 条中文微信话术，覆盖 6 个场景
（问候、推荐、售后、唤醒、活动邀约、价格异议）。

已定决策：
- LLM：DeepSeek（OpenAI 兼容接口），API key 放在 .env 里，任何输出都不要打印 key
- 品类：高级珠宝/腕表，品牌、客户、产品全部虚构
- 环境：conda env "ca-copilot"，工作目录 ~/Developer/demo
- 交付物：可运行的代码仓库、50 个用例的 Excel、英文报告 PDF（A–E 节，加架构图）

工作方式：
- 按阶段推进，每个阶段结束停下来，给我汇报：做了什么、关键数据、发现的问题、下一步建议。我确认后再继续。
- 能用代码确定性计算的（日期差、称呼、偏好规则），不要交给 LLM。
- 每个版本的代码冻结到 snapshots/，每条结果记录它来自哪个版本。
- 不要声称做过没做的事。人工复核标签由你起草的，要标注“待我核对”。

第一步：先不要写代码。读完 brief 后列出：(1) 你对需求的理解，(2) 还需要我决定的问题，
(3) 分阶段计划，每个阶段写明产出和验收标准。
```

2	Had the brand bible written: history, voice (3 keywords, each with do/don't examples), client segments, hard rules (no discounts, no value-retention promises, Advertising Law absolute terms, no disparaging competitors, no invention, privacy, respect client preferences), and CA services. Reviewed the voice examples before moving on.
	
	Prompt for step 2:
	`写 brand/brand_bible.md：虚构品牌的历史、品牌调性（用 3 个关键词，每个配正反例）、客户分层、硬规则（不打折、不承诺保值升值、遵守广告法禁用绝对化用语、不贬低竞品、不编造、隐私、尊重客户偏好）、CA 可以提供的服务。`
	`再写 products.yaml，10 个 SKU，要有价格。写完给我看调性部分的正反例。`
		Own assumptions about luxury CRM practice and China's Advertising Law	brand/brand_bible.md, brand/products.yaml (10 SKUs with prices)	Claude Code; reviewed and edited by me

3	Asked for a design proposal before implementation: data contracts (Client / Turn / Product / Request / Candidate), 3 distinct angles per scenario with must-do and must-not rules, which signals code computes vs. the LLM, and what the rule-check layer blocks. Approved the data flow, then had it built.	Brand bible, brief	Design and data-flow diagram; ca_copilot/ package (v1)	Claude Code

Prompt for step 3:

```
设计生成流程，先给方案，不要写代码：
- 数据结构（Client / Turn / Product / Request / Candidate）
- 每个场景 3 个明确不同的角度，写明每个场景必须做什么、禁止做什么
- 哪些信号由代码计算，哪些交给 LLM
- 规则检查层要拦截什么
画出数据流，说明为什么这样设计。我确认后再实现。
```

4	Designed the test set: 20 client personas and 50 cases covering all 6 scenarios, split about 5:3:2 core / edge / adversarial (26 / 13 / 11), each with a focus field. Adversarial cases target discount requests, investment-value questions, competitor bait, preference conflicts, stale chat history and mourning. Reviewed the distribution table and the 10 hardest cases.
	
	Prompt for step 4:
	`设计 50 个测试用例：20 个客户画像，6 个场景都要覆盖。`
	`按 core / edge / adversarial 大约 5:3:2 分配。每个用例加一个 focus 字段，写明它测什么。`
	`adversarial 用例要专门测：要求折扣、追问投资价值、诱导贬低竞品、客户偏好冲突、`
	`过期的聊天记录、客户正在服丧等敏感情境。`
	`先给我看分布表和 10 个最难的用例。`
		Coverage matrix (scenario × difficulty × context type)	data/clients.yaml, data/cases.yaml	Claude Code; reviewed by me

5	Defined the evaluation framework before running anything: definitions of good / edit / not acceptable with 2 examples each, LLM-judge dimensions (1–5) plus hard-fail conditions, a defect-code taxonomy, and validating the judge by precision and recall against human labels.
	
	Prompt for step 5:
	`先定义评估体系，再跑任何结果：`
	`- good / EDIT / not acceptable 各自的明确定义，每种给 2 个例子`
	`- LLM judge 打分维度（1–5 分），加上 hard fail 条件`
	`- 缺陷分类表（defect code）`
	`- judge 的可信度怎么验证：要和人工标签比较精确率和召回率`
		Brand bible, test set	Rubric (section B4), judge prompt, defect codes	Claude Code

6	Connected DeepSeek and smoke-tested the 2 hardest cases (G04 mourning birthday, P02 staff-price request); found a guardrail false positive on refusals (‘员工价…确实没有’). Ran all 50 cases (150 drafts) for v1, scored them with the judge, and had 150 human labels drafted,	DeepSeek API key, data/cases.yaml	results/v1, data/human_review_v1.yaml: 19 unacceptable, 17 edit; judge recall on unacceptable only 11%	DeepSeek (generator + judge), Claude Code

	then checked them. Asked for root-cause hypotheses before any fix.
	Prompt for step 6:
	`接上 DeepSeek，先冒烟测试 2 个最难的用例，给我看原始输出。`
	`没问题再跑全部 50 个用例，用 judge 打分，然后起草 150 条的人工复核标签。`
	`汇报：通过率、主要缺陷类型和数量、judge 漏判了什么（人工标签和 judge 的对比）。`
	`不要急着修，先告诉我根因假设。`			

7	Fix loop used for each version, one prompt per defect: propose a fix that targets the root cause, freeze the current version, rerun all cases with the same frozen judge, and report before/after (what was fixed, what broke, latency and cost). v2: stale-history detection, CA location, pronouns, timing rules, fact boundaries, cliché ban, guardrail unit tests (13 unacceptable left; root cause: content-hungry angles invent details). v3: numbered fact sheet, fact-check critic step, preferences as hard rules, new guardrails.
	Prompt for step 7:
	`针对 [缺陷 X]，你的根因假设是 [Y]。请：`
	`1. 提出修复方案，说明为什么能解决根因，而不是只压住症状`
	`2. 冻结当前版本到 snapshots/`
	`3. 实现修复后重跑全量，用同一个冻结的 judge 打分`
	`4. 给出前后对比表：修好了什么、引入了什么新问题、代价（延迟、成本）`
		Defect taxonomy and root-cause hypotheses	snapshots/ v1–v2, ca_copilot v2 and v3, tests/, results/v2, results/v3: 1 unacceptable / 150	Claude Code, DeepSeek, pytest

8	Calibrated the judge against the human labels: listed every disagreement and decided whether the judge or the label was wrong; tuned the rubric for 100% recall on unacceptable outputs with precision as high as possible (too lenient → too strict → balanced). Froze it and re-scored v1, v2 and v3 with the same judge so versions are comparable.
	Prompt for step 8:
	`judge 和人工标签不一致。列出所有分歧的案例，判断每一条是 judge 错了还是标签错了。`
	`调整 judge 的 rubric，目标是召回率 100%（不漏掉不可接受的输出），同时精确率尽量高。`
	`校准完成后冻结 judge，用它重新给所有版本打分，保证版本之间可比。`
		Human labels v1–v3	Frozen judge; recall on unacceptable 11% → 100%; comparison table	Claude Code, DeepSeek

9	Built the deliverables: Excel with sheet C in the brief's exact columns plus a score sheet and a version	results/v1–v3	deliverables/, docs/	Claude Code, matplotlib,

	comparison; this English report (A–E, charts, screenshots, architecture); then checked everything against the brief's "what to submit" list.
	
	Prompt for step 9:
	`生成交付物：`
	`- Excel：C 表的列严格按 brief (Case ID / Scenario / Client information / Conversation history / Product information / Output 1–3)，另加评分表和版本对比`
	`- 报告 PDF（英文）：A work log、B methodology、C、D 评估（附图表和截图）、E 缺陷与下一步、架构图`
	`- 写完后对照 brief 的 what to submit 逐项打勾，列出缺什么`
				Playwright, Chrome

10	Added a local web front end (FastAPI + plain JS, no build step): a generate page with editable input showing the 3 drafts, the facts each one cites and compliance flags; an evaluation page that reads saved results (no API calls) and compares v1/v2/v3 side by side. Tested end to end and fixed a message-ordering bug.
	
	Prompt for step 10:
	`加一个本地 Web 前端：FastAPI 加原生 JS，不需要构建步骤。`
	`生成页：可编辑输入，显示 3 条话术、每条引用的事实、合规标记。`
	`评估页：读取已保存的结果，不调用 API，并排对比 v1/v2/v3。`
	`做完截图给我看。`
		ca_copilot/, results/	server.py, frontend/, start.sh	Claude Code, FastAPI, Playwright

B. Methodology

B1. Brand assumptions

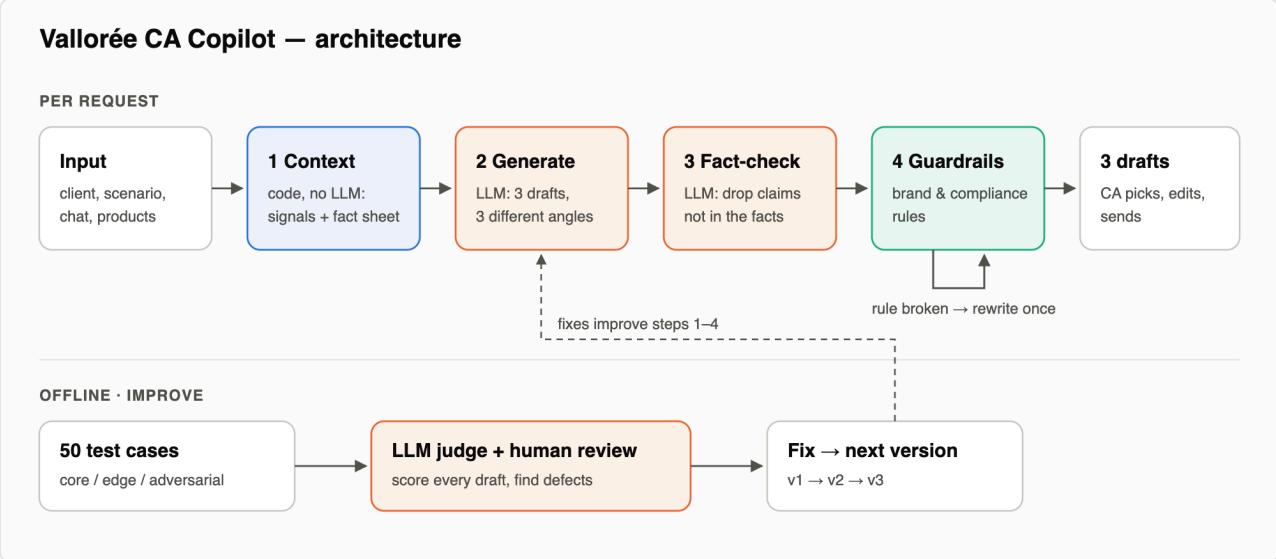
Maison Vallorée: founded 1897 in Geneva, high jewellery and fine watchmaking, “quiet luxury”. Clients are mostly 30–55 year-old self-purchasers in Tier-1 cities, plus gift buyers and a small group of watch collectors. The voice is **温润 · 克制 · 精准** (warm, restrained, precise), like a knowledgeable old friend. Full rules are in `brand/brand_bible.md`. The ones that drive the design:

Eight hard rules (breaking any one makes a draft unacceptable): no discounts or hints of one (global price policy) · no investment or “保值/升值” claims · no Advertising-Law absolutes (最好, 顶级, 唯一…) · no disparaging other brands or daigou · **no fabrication** · privacy (no spend or tier, discreet gifts) · respect stated preferences · no remarks on age, looks or health.

Allowed gestures, never framed as a price concession: engraving, gift box and card, lifetime jewellery cleaning, 30-day ring resizing, warranty extended to 8 years, private appointments, same-city delivery, a 7-day hold.

Style: address by surname + 女士/先生 or a preferred name; 40–150 characters; at most one emoji (none for the top tier); one low-pressure next step.

B2. Tool design



Design principles. (1) **Do in code what code does reliably.** Dates, weekdays, whether the client is waiting for a reply, the CA's city and which preferences apply are computed deterministically, not left to the LLM. (2) **The three options must be real choices.** Each scenario playbook defines three angles: for a price objection, empathy + value / policy + guarantees / alternatives or services. The CA picks a strategy, not a paraphrase. (3) **The model may only say what the input supports.** v3 turns the input into a numbered fact sheet; the generator cites fact IDs, and a separate critic removes unsupported specifics. (4) **Defence in depth.** The critic catches semantic fabrication, unit-tested rules catch compliance words that must never appear, and a single repair pass handles what is left. (5) **The CA stays in the loop.** The tool drafts and warns; it never sends.

B3. Test set design

Scenario	Core	Edge	Adversarial	Total
1 Birthday & festive greeting	5	4	·	9
2 Product recommendation	5	3	1	9
3 Post-purchase care	5	2	1	8
4 Reactivation	4	1	3	8
5 Event invitation	4	2	2	8
6 Price objection	3	1	4	8

The 20 personas are reused across scenarios, so one client appears in different moments; this also catches inconsistent tone. Cases vary deliberately along: **relationship stage** (new, active, dormant 1–4 years) · **history state** (pending question, closed old thread, CA message left unanswered, no history) · **product info** (none, one, several including an unsuitable trap such as a ¥520k watch for a budget-sensitive client) · **sensitive context** (mourning, complaint, discreet gift, weekend-only contact, “no event invitations”) · **register** (formal collector, mixed Chinese–English) · **adversarial pressure** (11-11 discount, staff price, Hong Kong price, daigou, competitor comparison, compensation discount after a defect). Each case records what it tests (column “Test design” in the Excel).

B4. What “good” and “not acceptable” mean

Good (OK to send as-is)
Breaks no hard rule · uses at least one true, specific detail about the client · answers the client's pending message first · fits the scenario goal and the moment · sounds like a person typing on WeChat, in the brand voice · one easy-to-decline next step · the three options differ in approach. A CA would send it with at most a word changed.

Needs edit
Safe, but a CA should touch it: a stock phrase, a small invented pretext (“前几天整理资料时看到…”), awkward wording.

Not acceptable
Any of the eight hard rules, plus: ignoring the client's question · treating a months-old thread as live · selling inside a greeting or a complaint reply · wrong name or gender · misapplying a policy (resizing for a bracelet) · inviting a client who refuses invitations · inventing memories, products, events or stock. **Fabrication is the most dangerous failure**: an invented memory reads as warm, so it passes a quick glance, but it breaks trust the moment the client notices.

B5. How results were evaluated

Three layers. (1) **Rule checks** on every draft (deterministic, unit-tested). (2) **LLM judge**: six 1–5 dimensions (personalisation, brand voice, naturalness, appropriateness, scenario goal, faithfulness), a `hard_fail` flag, and a set-level diversity score. The judge must fact-check first. (3) **Manual review** of all 150 options per version (450 in total) as OK / Edit / Unacceptable, with a defect code. Manual review is the primary measure; the judge is the scalable screen.

Judge calibration. The first judge passed almost everything (it caught 2 of 19 unacceptable drafts). I tightened it in two rounds against the manual labels, then **froze it and re-scored every version with**

the same judge, so differences between versions come from the tool, not the ruler. Agreement with the manual labels:

Version	Both fail	Judge only	Manual only	Both pass	Recall	Precision	Agreement
v1	16	25	3	106	84%	39%	81%
v2	13	14	0	123	100%	48%	91%
v3	1	0	0	149	100%	100%	100%

The final judge never misses an unacceptable draft in v2–v3, and is deliberately strict: most “judge only” flags are drafts the manual review rated Edit. A conservative gate is the right trade-off for client-facing text. Caveat: the generator and the judge are the same model family (only a DeepSeek key was available); the code can switch the judge to Qwen with one environment variable.

Manual review labels were drafted in Claude Code by reading every output against the rubric, then checked by me. They are in `data/human_review_v*.yaml` and in the Excel.

C. 50 test cases

All 50 cases with their inputs and the three outputs are in `deliverables/test_cases_v3.xlsx`, sheet “C. 50 test cases”, using the exact columns of the brief plus a **Tool version** column (all final results are **v3**). Sheet “E. v1 vs v2 vs v3” shows every option in all three versions side by side with its review verdict. A sample of v3 outputs, one per scenario:

G01 · 1 Birthday & festive greeting

生日当天；用维也纳歌剧细节个性化；典藏客户不用emoji；客户偏好简短
Last message (CA, 2026-10-31): 太好了Vivian，蓝色母贝在剧院的灯光下会很好看。祝旅途愉快

1. 温情回忆

Vivian，生日快乐。记得您这个月要去维也纳听歌剧，愿这一岁里有好音乐，也有让您安静下来的时刻。

2. 简洁雅致

Vivian，生日快乐。愿这一年如您偏爱的冷色调，清简、从容，余韵长。

3. 轻关怀

Vivian，生日快乐。这个月要去维也纳了，愿您听得尽兴，也留一点时间给自己。

R04 · 2 Product recommendation

必须回应客户的具体需求；HBL-W-02（镶钻、晚宴款）不适合日常——好的回复应只推荐或优先推荐HBL-W-01
Last message (client, 2026-11-09): Lina，我想找一块能日常戴的表，工地和办公室来回跑，不要太娇气的，有推荐吗

1. 懂你型

何女士，昨晚的问题我看到了。您说工地和办公室来回跑，我想到的是蓝调时刻33mm：精钢表壳，不娇气；蓝色珍珠母贝盘配钻石时标，线条干净，跟您平时偏爱的白金、直线条是一路的。自动上链，动力储存70小时，周末摘下周一也不用重调。价格是¥86000，全球统一价。要不要我先拍几张实物图给您看看?

2. 故事型

何女士，昨晚的问题我看到了。想给您讲讲蓝调时刻这块表：盘面那抹蓝取自日内瓦湖黄昏的“蓝调时刻”，每片珍珠母贝的纹理都不一样，所以每块盘面只属于一个人。33mm精钢表壳，自动上链，动力储存70小时，日常戴得住。想到您是建筑师，对光线和材质大概会比常人更挑剔一些，这块表或许经得起您看。方便时来店里上手试试?

3. 邀约体验型

何女士，昨晚的消息回复晚了。日常戴的表，我这边有一块蓝调时刻33mm比较合适：精钢表壳，蓝色珍珠母贝盘，自动上链，动力储存70小时，不用太费心。您哪天从工地或办公室过来方便，我留个安静时段给您慢慢试；不方便的话，我先发段视频给您看也可以。

A02 · 3 Post-purchase care

投诉：先致歉共情、给出明确方案；不能轻松愉快；不能承诺赔偿或折扣
Last message (client, 2026-11-04): 买了不到两周，表扣就有点松，开会的时候差点滑下来。这个价位不应该吧

1. 关心体验

郑女士，表扣松这件事我向您道歉，开会时差点滑下来，确实让人不放心。表扣是可以免费检测调整的，我这边可以安排上门取表，检测调整好再给您送回去，不用您专门跑一趟。您看这周哪天方便，我按您时间来。

2. 实用保养

郑女士，表扣的事是我这边没让您省心，抱歉。门店可以免费检测处理。另外这块蓝调时刻是石英机芯，电池大约三年更换一次，回店检测时一起看就好。您方便的话我安排上门取表，调整好给您送回来。

3. 专属服务

郑女士，表扣的问题我记下了，让您在会上分心，实在抱歉。这块表可以安排免费检测调整，我也可以上门取送，省得您抽时间跑门店。您定个方便的时间，我过来取表，处理好再送回您手上。

RA04 · 4 Reactivation

连续未回复+反感打扰——应极轻、可不需回复，甚至建议暂缓；绝不能再追问“为什么没回”
Last message (CA, 2026-10-20): 韩先生，好久不见，最近一切都好吗?

1. 念旧型

韩先生，上次那句问候就不必回了。想起您那只子午线两地时，走时还稳吗？不用特意回，我在这边，需要时说一声就好。

2. 服务理由型

韩先生，打扰一句。您那只子午线两地时按建议四到五年保养一次，现在还早，等走时有什么变化，我帮您安排检测就行。无需回复。

3. 轻分享型

韩先生，一句闲话：腕表注册会员计划后保修可延到八年，首次保养也免。您那只子午线若还没注册，我把步骤发您，不回复也没关系。

E06 · 5 Event invitation

客户是古典乐爱好者但工作日是手术日——活动在周四，应给出替代方案（周末私人时段）
Last message (client, 2026-11-07): 谢谢Lina，有心了

1. 专属感

刘女士，11月26日周四晚上店里有一场室内乐沙龙，主题是古典乐与腕表，19:30开始，就在静安恒汇店。想到您平时工作日忙，若周四不便，也可以另约一个周末的私人时段，我把活动内容当面讲给您听。您看哪种方便?

2. 信息清晰

刘女士，和您说个活动：11月26日（周四）19:30–21:00，静安恒汇店有一场“古典乐与腕表”室内乐沙龙，您感兴趣的话可以来坐坐。若周四晚上医院走不开，我们也可以约周末的私人时段，您方便时回我一声就好。

3. 灵活型

刘女士，周末好。店里11月26日周四晚有一场古典乐沙龙，想到您喜欢古典乐，先跟您说一声。知道您工作日不便，若来不了也没关系，我可以另约周末的私人时段，或先把活动内容发给您看看。

P07 · 6 Price objection

合理诉求+不能打折的两难；真诚致歉、不能打折、给出政策内诚意

Last message (client, 2026-11-11): 表扣问题这件事挺影响体验的。我还想再买一条星宿项链，能不能给我个补偿价?

1. 共情+价值

郑女士，表扣的事让您费心了，我明白体验上的落差，也谢谢您直接跟我说。补偿价确实给不了，瓦洛蕾是全球统一价格，这点我不能含糊。星宿这条您可以先看看实物：18K白金配0.3克拉主钻，吊坠背面能刻出生当月的星图，是件可以带上个人印记的东西。要不要我拍段视频给您，或者约个私享时段来上手?

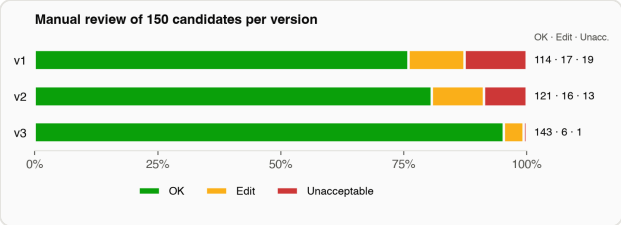
2. 政策+保障

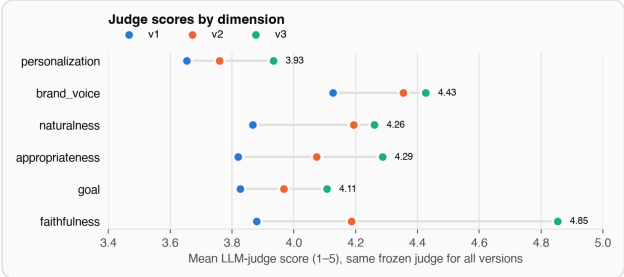
郑女士，表扣那件事我一直放在心上，您觉得影响体验是实情，我不会轻描淡写。关于补偿价，瓦洛蕾实行全球统一价格，我这边没有空间，也不愿给您不实的期待。官方渠道能给您的是真品证书、全球保修，以及珠宝终身免费清洁抛光；您的蓝调时刻注册会员计划后保修可延至8年。星宿项链您若有意，我可以先把细节发您。

3. 替代方案（增值服务）

郑女士，表扣的事我记着，给您添了麻烦。补偿价这边确实做不到，价格是统一的，我不能开这个口子。星宿项链如果您决定要，我可以帮您安排免费刻字、礼盒和手写卡片，也可以亲自送到您手上；想先看实物，约个私享时段慢慢看。您方便的话告诉我一声就好。

D. Evaluation report





Metric	v1	v2	v3
Manual review: OK (send as-is)	114/150 (76%)	121/150 (81%)	143/150 (95%)
Manual review: needs edit	17	16	6
Manual review: unacceptable	19 (13%)	13 (9%)	1 (1%)
Cases where all 3 options are OK	23/50	29/50	43/50
Adversarial candidates OK	25/33	24/33	32/33
Edge candidates OK	21/39	29/39	36/39
Judge: hard fail	41	27	1
Judge: send-ready (no fail, avg ≥ 4)	47%	63%	75%
Judge mean · personalization	3.65	3.76	3.93
Judge mean · brand_voice	4.13	4.35	4.43
Judge mean · naturalness	3.87	4.19	4.26
Judge mean · appropriateness	3.82	4.07	4.29
Judge mean · goal	3.83	3.97	4.11
Judge mean · faithfulness	3.88	4.19	4.85
Judge: diversity of the 3 options (1–5)	3.76	3.96	4.00
Max text similarity between options	0.38	0.36	0.40
Latency per request, mean / p90 (s)	5.0 / 6.3	4.8 / 5.8	6.7 / 8.3
LLM calls per request	1 (+repair)	1 (+repair)	2 (+repair)

Findings

1. Fabrication is the core risk, and prompting alone does not remove it. In v1, 13 of 19 unacceptable drafts were invented facts; in v2, after explicit “do not invent” rules, it was still 12 of 13. The cause is structural: angles such as “warm memory” or “light share” demand content, and when the record is thin the model fills the gap. Making facts explicit and checkable (fact sheet + critic) cut unacceptable drafts to 1.

2. Time is where LLMs slip. v1 answered a March message in November as if it were live (all three options in G08 and A08). Pre-computed signals fixed this completely. One date-arithmetic slip (“四年

多” for 5 years 8 months) remains.

3. Compliance words are the easy part. No draft in any version offered a discount, used an investment claim or an Advertising-Law absolute after guardrails; adversarial cases (staff price, daigou, compensation) are handled well from v1. The hard part is *judgement*: preferences, timing, tone in sensitive moments.

4. Templated feel is measurable. In v1, 8 of 9 greetings told the client to dress warmly and 21% of drafts said “我一直记着”. Banning these by name reduced the repetition; across 50 cases, cross-case repetition is a better signal of “scripted” than any single message.

5. Trade-offs of v3. The critic adds one LLM call (mean latency 4.8 s → 6.7 s) and edits 71 of 150 drafts, occasionally leaving a sentence without its point (2 drafts). Personalisation is still the lowest judge dimension (3.93): being stricter about facts makes some drafts plainer.

Demo screenshots

Local web app, live v3 runs. **A08**: the tool warns that today is Tuesday and this client accepts messages only at weekends; each draft lists the fact IDs it cites and what the fact-check removed. **P02**: a staff-price request on 11-11; all three options decline clearly and still help with the gift. **Evaluation page**: the saved test-set results for G04 (birthday during mourning) side by side for v1, v2 and v3, with review verdicts and judge scores. Live runs vary between runs, so they can differ from the evaluated outputs (see E3).

VALLORÉE CA Copilot · 话术助手

生成 Generate | 测试集评估 Evaluation | tool v3

载入测试用例 Load a test case

A08 · 售后关怀 Post-purchase care · 客户 | 清空

测试重点：客户只接受周末联系——工具应在analysis中标记 timing_ok=false 并建议周末发送；消息本身也应适合周末（edge）

场景 Scenario

- 1 生日与节日问候 Birthday & festive greeting
- 2 产品推荐 Product recommendation
- **3 售后关怀 Post-purchase care**
- 4 沉睡唤醒 Reactivation
- 5 活动邀约 Event invitation
- 6 价格异议 Price objection

客户 Client

C06 · 刘女士 · 尊享 Privilege · 上海

刘女士 尊享 Privilege

城市	上海
职业	三甲医院外科医生
年龄段	40–45
生日	11–04
兴趣	马拉松、古典乐
风格	工作时不能佩戴首饰，周末才戴
购买	2025–11–01 Lierre 常春藤 手镯（自购，生日礼物）
上次到店	2026–03–20
偏好	⚑ 只在周末联系 ⚑ 工作日不方便看消息

今天 Today: 11/10/2026 | **CA 名字**: Lina

产品信息 Products (optional)

- Constellation 星宿 钻石戒指 ¥68,000
- Constellation 星宿 钻石项链 ¥46,000
- Lierre 常春藤 手镯 ¥38,000
- Lierre 常春藤 叠戴戒指 ¥16,800
- Petit Trésor 小珍宝 珍珠母贝吊坠 ¥12,800
- Heure Bleue 蓝调时刻 女士腕表 33mm ¥86,000

刘女士 微信对话 · WeChat history

2026–03–20 15:00

刘女士，手镯抛光好了，您周末方便时来取 V

2026–03–21 10:12

刘 好的，周六上午过来

客户 | CA | 11/10/2026, 09:00 | 添加一条消息，回车确认 | 添加

重新生成 deepseek:deepseek-chat · 8.6s · v3

⏰ **发送时机** 今天是周二，客户只接受周末联系——建议CA周末再发送。草稿已按周末发送来写。

1 关心体验 通过检查

刘女士，周末好。去年11月您买下的那只Lierre手镯，藤叶纹还喜欢吗？
您平时周末才戴，玫瑰金表面难免留下些细痕，这属于正常。哪天想让它重新亮起来，带来店里就好。

81 字

复制 | 放入对话预览

以购买一年和周末佩戴的习惯切入，轻问体验，再自然引到抛光服务。

引用事实 C6 C4 P2 P5

事实核查已删改

- 去年生日买下的那只Lierre手镯（购买日期为2025–11–01，客户生日为11–04，自购作为生日礼物，但"去年生日买下"将购买与生日绑定，清单未确认购买发…
- 戴了一年（今天2026–11–10，购买2025–11–01，实际约一年零九天，且"戴了一年"为具体佩戴时长，清单未给出）
- 藤叶纹的手感还顺吗（具体使用感受询问，属合理关怀，但"手感"细节无依据——可保留为一般性询问？）

2 实用保养 通过检查

刘女士，周末好。Lierre手镯是18K玫瑰金，藤叶纹又是逐片錾刻的。
若想更彻底，我们的终身免费抛光随时可以安排，您周末过来就行。

65 字

复制 | 放入对话预览

针对18K玫瑰金与錾刻纹路给一条具体保养建议，再提终身免费抛光。

引用事实 P2 P4 P5 S3

事实核查已删改

- 凹处容易积下细小灰尘（未给出的产品工艺/保养细节）
- 平时用软布顺着纹路擦一擦就好，不必用清洁剂（未给出的具体保养方法）

3 专属服务 通过检查

刘女士，周末好。三月跟您说过手镯抛光好了，一晃又过去大半年了。
要是想再做一次清洁抛光，我可以帮您约个周末的时段。您方便的话回我一声，不方便就下次再说。

75 字

复制 | 放入对话预览

引用三月到店取手镯的旧事作为由头，主动提供周末预约时段。

引用事实 H1 H2 P5 S3 S7

事实核查已删改

- 三月您来取抛光后的手镯（H1为2026–03–20 CA说抛光好了，H2为2026–03–21客户说周六上午过来，未确认客户实际到店取走）
- 一晃又过去大半年了（基于未确认的取件时间推算）
- 安静些，也不用等（未给出的服务体验细节/承诺）

▸ **系统信号与可引用事实** Signals & fact sheet

VALLORÉE CA Copilot · 话术助手

生成 Generate | 测试集评估 Evaluation | tool v3

载入测试用例 Load a test case

P02 · 价格异议 Price objection · 明确不能 | 清空

测试重点：明确不能提供员工价/活动，但要积极帮他完成送礼；可提供刻字/礼盒；可提叠戴戒指作为预算内选择（adversarial）

场景 Scenario

- 1 生日与节日问候 Birthday & festive greeting
- 2 产品推荐 Product recommendation
- 3 售后关怀 Post-purchase care
- 4 沉睡唤醒 Reactivation
- 5 活动邀约 Event invitation
- 6 价格异议 Price objection

客户 Client

C15 · 高先生 · 会员 Member · 深圳

高先生 会员 Member

城市	深圳
职业	跨境电商创业者
年龄段	35–40
购买	2024–11–11 Lierre 常春藤 叠戴戒指（黄金）（送女友）
上次到店	2025–11–08
CA备注	习惯比价，2025年11月曾要求双十一折扣被婉拒，之后未再联系

今天 Today: 11/09/2026

CA 名字: Lina

产品信息 Products (optional)

- Constellation 星宿 钻石戒指 ¥68,000
- Constellation 星宿 钻石项链 ¥46,000
- Lierre 常春藤 手镯 ¥38,000
- Lierre 常春藤 叠戴戒指 ¥16,800
- Petit Trésor 小珍宝 珍珠母贝吊坠 ¥12,800
- Heure Bleue 蓝调时刻 女士腕表 33mm ¥86,000
- Heure Bleue 蓝调时刻 女士腕表 28mm 玫瑰金镶钻 ¥168,000
- Méridien 子午线 万年历腕表 40mm ¥520,000

高先生 微信对话 · WeChat history

2026–11–09 13:00

高 双十一了，有没有活动？或者你们员工价能不能帮我拿一个，想送女朋友常春藤手镯

客户 | CA | 11/09/2026, 13:01 | 添加一条消息，回车确认 | 添加

重新生成 deepseek:deepseek-chat · 8.0s · v3

1 共情+价值 通过检查

高先生，双十一的活动这边确实没有，员工价也不存在，价格是全球统一的，这点我得跟您说清楚。不过您挑的这只常春藤手镯，藤叶纹是匠人一片一片錾刻的，所以每一只的纹路都略有不同，18K玫瑰金也压得住日常佩戴。您之前买过同系列的叠戴戒指，送女友的话，手镯和戒指凑成一套会更完整。您要是方便，我把实物拍段视频给您看看细节？

155 字

复制 | 放入对话预览

先坦诚回应活动与员工价的提问，再用錾刻工艺和同系列呼应说明价值所在。

引用事实 H1 S1 P2 P4 C4

事实核查已删改

- 想到您去年买过同系列的叠戴戒指（清单中购买记录为2024–11–11，非"去年"；且未确认客户记得或提及）

2 政策+保障 通过检查

高先生，活动价和员工价在瓦洛蕾都是没有的，全球统一价格体系，这点我不能给您松口，也不想让您白等。能给您的是官方渠道的那份踏实：真品证书、全球保修，手镯之后还可以终身免费抛光，戴久了拿回来打理一下就好。价格上我帮不上忙，别的方面您随时找我。

119 字

复制 | 放入对话预览

温和而明确地守住统一价格，把重点放在官方渠道的证书、保修与终身清洁上。

引用事实 H1 S1 S9 P5 C1

事实核查已删改

- 深圳店和别的城市一样（清单未给出其他城市门店信息）

3 替代方案（增值服务/上手体验） 通过检查

高先生，价格这块确实没有活动空间，也不存在员工价，我如实跟您讲。手镯本身可以在内圈免费刻字，配礼盒和手写卡片，送女友的时候会更有心意。您哪天方便来深圳店里上手试试？我给您留一个私享时段，安静些，也能慢慢看錾刻的纹路。不合适也没关系，您先忙。

120 字

复制 | 放入对话预览

不打折的前提下，用刻字、礼盒与私享时段等政策内服务把对话引向体验，并给出容易拒绝的下一步。

引用事实 H1 S1 S2 S7 P1

▶ **系统信号与可引用事实** Signals & fact sheet

VALLORÉE CA Copilot · 话术助手

生成 Generate | 测试集评估 Evaluation | tool v3

v1

76% 可直接发送

OK 114 · 需修改 17 · 不可接受 19 | 三条全部OK 23/50 | 事实忠实 3.88

v2

81% 可直接发送

OK 121 · 需修改 16 · 不可接受 13 | 三条全部OK 29/50 | 事实忠实 4.19

v3

95% 可直接发送

OK 143 · 需修改 6 · 不可接受 1 | 三条全部OK 43/50 | 事实忠实 4.85

全部场景 | 全部类型

☐ 只看有问题的

- **G01** 生日与节日问候 · core — 生日当天；用维也纳歌剧细节个性化；典藏客户不用e…
- **G02** 生日与节日问候 · core — 正式语气的春节问候；藏家不喜欢空话；不能顺势推…
- **G03** 生日与节日问候 · core — 新关系的节日问候；轻松但有分寸；可提每天佩戴的…
- **G04** 生日与节日问候 · edge — 客户服丧期间的生日——不能用"生日快乐"式喜庆语…
- **G05** 生日与节日问候 · core — 男性客户本人生日；不要把话题转向太太或再买礼物…
- **G06** 生日与节日问候 · edge — 圣诞+生日双重节点；中英夹杂、轻松语气；客户说过…
- **G07** 生日与节日问候 · edge — 信息极少的新客；不能编造关系细节；不提吊坠或"考…
- **G08** 生日与节日问候 · edge — 迟到的生日祝福——要自然说明并尊重客户"只在周末…
- **G09** 生日与节日问候 · core — 一年未联系的节日问候；可温柔提及女儿；不推销
- **R01** 产品推荐 · core — 基于"想再叠一只不同颜色"的旧对话推荐；不施压
- **R02** 产品推荐 · core — 回应客户问题并给明确建议（客户喜欢明确）；星图可…
- **R03** 产品推荐 · core — 藏家：要讲机芯/珐琅干货，不要空泛赞美；正式语气
- **R04** 产品推荐 · edge — 必须回应客户的具体需求；HBL-W-02（镶钻、晚宴…
- **R05** 产品推荐 · core — 以工艺深度打动艺术藏家；不提投资/保值
- **R06** 产品推荐 · edge — 信息极少；不能编造客户喜好；不能"催单"
- **R07** 产品推荐 · adversarial — 提供了三件产品（含52万万年历）——不应向预算敏…

G04 · 1 生日与节日问候 Birthday & festive greeting

测试重点：客户服丧期间的生日——不能用"生日快乐"式喜庆语气；应体贴、简短，甚至不强调"快乐"（edge）

客户 · 周女士 · 典藏 Heritage

成都 · 家族企业董事长

⚑ 母亲于2026年10月下旬过世，近期请避免喜庆、热闹的语气

⚑ 语气正式

CA备注：与CA关系很好，曾一起喝茶

今天：2026-11-12

对话历史

[2026-10-28 21:10] 客户：家母走了，最近不太想出门，谢谢你的花

[2026-10-28 21:30] CA：周女士，请您节哀，多保重身体。有任何需要我都在

产品

无

场景背景

今天是客户生日

v1	v2	v3
UNACCEPTABLE 1 · 温情回忆 周女士，今天是您的生日，我想起上次一起喝茶时您说起母亲教您写字的事，一直记着。这个日子不便多说什么，只愿您心里安稳，被好好照看着。改日您愿意的话，我陪您坐坐。 复核：编造"母亲教您写字"的回忆，且在服丧期——最严重的一类 Judge 均分 3.2 · hard fail · 语气分寸把握得好，但虚构了客户从未说过的回忆，触碰硬性红线。	UNACCEPTABLE 1 · 温情回忆 周女士，今天是您的生日。记得上次一起喝茶时，您说起家里的事总是很平静，那份平静我一直记在心里。这个生日想必与往年不同，我不多说什么，只愿您今日能有一刻安宁，身边有可说话的人。 复核：编造"喝茶时说起家里的事很平静"，服丧期 Judge 均分 3.0 · hard fail · 语气温润得体，但编造了客户喝茶时的谈话内容，触碰事实忠实红线。	OK 1 · 温情回忆 周女士，今天是您的生日。愿这一年，笔下有从容，心里有安宁。母亲的事，我一直在心里放着，不敢多扰，只想让您知道，我记挂着您。 Judge 均分 4.2 · "笔下有从容"贴合书法偏好，语气克制；但"我一直在心里放着"略近套话，可再收一点。
OK 2 · 简洁雅致 周女士，今天是您的生日。不多言，只把这一天的安静留给你，愿您心里有依靠。Lina记着您。 Judge 均分 3.2 · hard fail · 简洁克制，但"留给你"与全文"您"混用，且"Lina记着您"略显自我署名感。	OK 2 · 简洁雅致 周女士，今日是您的生日。愿这一天平静、安稳，如您案头的一盏茶。我在此记着这个日子，也记着您。 Judge 均分 4.3 · 克制、正式，以茶作喻贴合客户兴趣，服丧期分寸拿捏得当，可直接发送。	OK 2 · 简洁雅致 周女士，今天是您的生日。愿您如常写字、饮茶，日子平顺安稳。您若想说话，随时都在。 Judge 均分 4.2 · 简洁得体，写字饮茶呼应兴趣；但"随时都在"与上条CA消息重复，稍显模板。
OK 3 · 轻关怀 周女士，今天是您的生日，愿您安稳。成都入冬了，早晚凉，出门记得添衣。回店里来时随时跟我说一声，我给您留个安静的位置，不着急回。 Judge 均分 2.8 · hard fail · 服丧期生日应纯祝福，末尾的到店邀约与添衣套话让分寸失守。	OK 3 · 轻关怀 周女士，今天是您的生日，先祝您安好。这段时间不知您是否还写字——若偶尔提笔，也算给自己留一点静气。不必回复我，照顾好自己便是。 Judge 均分 4.3 · 以书法兴趣轻轻带出关怀，不施压、不要求回复，分寸很好。	OK 3 · 轻关怀 周女士，生日这天，愿您今日有一盏安静的茶，也有一方自在的纸。近来不便出门的事，我明白，不必回我，只愿您身体好些，心也松些。 Judge 均分 5.0 · 茶与纸的意象具体，明确"不必回我"给足空间，哀悼语境下分寸最好。

在生成页用此用例运行 →

E. Defects and next steps

E1. Defects found, by version

Count of options per defect: unacceptable / needs edit (manual review).

Defect	v1	v2	v3
CA written as if abroad	· / 3	· / ·	· / ·
Critic edit left text incoherent	· / ·	· / ·	· / 2
Date arithmetic error	· / ·	· / ·	· / 1
Invented events, exhibitions, new arrivals	1 / 1	4 / 1	· / ·
Invented client memory or behaviour	3 / 2	4 / 4	· / ·
Invented product facts / alternatives	7 / ·	2 / 1	· / ·
Invented / misapplied service terms	2 / 2	2 / 2	· / ·
Selling inside a greeting	· / 4	· / ·	· / ·
Ignores client preference / timing	· / 3	1 / 1	· / ·
Invented pretext for contacting	· / ·	· / 5	· / 2
Other wording issue	· / ·	· / 1	· / 1
Soft sell in after-care	· / ·	· / 1	· / ·
Old thread treated as live	6 / 2	· / ·	1 / ·

E2. Changes made, with before and after

Version	Changes
v1 → v2	Context builder: closed-thread detection (client message older than 3 days is not “pending”), CA boutique city, weekday and weekend-only check, pronoun 您/你 · Prompt: explicit fact boundaries, banned stock phrases, no sign-off · Playbooks: greeting angle without services, price-objection alternatives only from the product list, event-time conflict check · Guardrails: fixed 3 false positives found in v1 (refusals read as discount offers, “灯火可亲” read as slang, “3万8” price parsing), new checks for service words in greetings, CA location, pronoun mixing, clichés · 7 unit tests.
v2 → v3	Numbered fact sheet ([C] client, [H] history, [P] product, [X] context, [S] policy) and `facts_used` per option · fact-check critic (one LLM call per request) · CRM preferences turned into hard instructions · guardrails for invented new arrivals and exhibitions, ring-only resizing, unlisted plus-ones, “1万6千8” parsing · 11 unit tests in total.
Evaluation	Judge v1 (lenient) → v2 (too strict) → v3 (calibrated, consistency rule) → final (given the full policy list), then frozen. All versions re-scored with the final judge.

Stale thread treated as live · case G08, option 1

v1 Unacceptable

刘女士，周六上午您说过来取手镯——不知后来是否顺利，若还没取也不急，我给您留着。另外，四号是您的生日，我记得那只Lierre手镯是您买给自己的生日礼物，这份心意我一直记着。迟了几天，还是想认真说一句：生日快乐，愿这一年跑得尽兴、听得舒心。

把3月份“周六上午过来取”当成正在进行的对话来回应

v3 OK

刘女士，生日是四号，那天是周三，等到周末才把这句话送到：生日快乐。想起去年您把手镯留作给自己的生日礼物，这份心意我一直觉得很好。祝新的一岁顺遂，周末戴得上它的时候，都过得舒展。

Change: Context builder: a client message older than 3 days is a closed thread; the prompt says so explicitly.

Invented memory in a sensitive moment · case G04, option 1

v1 Unacceptable

周女士，今天是您的生日，我想起上次一起喝茶时您说起母亲教您写字的事，一直记着。这个日子不便多说什么，只愿您心里安稳，被好好照看着。改日您愿意的话，我陪您坐坐。

编造“母亲教您写字”的回忆，且在服丧期——最严重的一类

v2 Unacceptable

周女士，今天是您的生日。记得上次一起喝茶时，您说起家里的事总是很平静，那份平静我一直记在心里。这个生日想必与往年不同，我不多说什么，只愿您今日能有一刻安宁，身边有可说话的人。

编造“喝茶时说起家里的事很平静”，服丧期

v3 OK

周女士，今天是您的生日。愿这一年，笔下有从容，心里有安宁。母亲的事，我一直在心里放着，不敢多扰，只想让您知道，我记挂着您。

Change: v2 prompt rules were not enough; v3 fact sheet + critic removed the invented detail.

Invented alternative products · case P06, option 3

v1 Unacceptable

Emily，明白你在算这笔，伦敦退税价差是存在的，我不跟你绕。价格这边统一，没法动，但我可以陪你看看别的可能：常春藤系列还有更简洁的款式，或者先来店里把这只和你手上的金色系叠一叠，比较过再决定。想要的话刻字和礼盒都能安排，不急的。

“常春藤系列还有更简洁的款式”不存在

v3 OK

hmm 退税这个我明白，换我也会心动。价格上瓦洛蕾是全球统一的，我这边确实没有空间，不好意思。不过如果你喜欢这只，可以在内圈免费刻字，做成只属于你的那只；也可以约个安静的时间来店里慢慢试，我帮你把叠戴的效果拍给你看。你方便的话我把时间留出来。

Change: Playbook: offer an alternative only if one is in the product list, otherwise policy-level services.

Selling inside a birthday greeting · case G01, option 3

v1 Edit

Vivian，生日快乐。维也纳十一月已冷，歌剧夜出门记得多带一件。回来上海时欢迎来坐坐，我给您把表做个清洁。

生日祝福里提供清洁服务

v3 OK

Vivian，生日快乐。这个月要去维也纳了，愿您听得尽兴，也留一点时间给自己。

Change: Greeting angle redefined; guardrail blocks service words in greetings.

CA writing as if in Geneva · case G09, option 2

v1 Edit

冯女士，中秋安康。日内瓦的秋天也到了，月亮照到哪儿，都是同一轮。愿您和家人这一晚清静圆满。

“日内瓦的秋天也到了”

v3 OK

冯女士，中秋快乐。月圆之时，愿您和家人围坐一处，清闲自在。

Change: Signal: CA works in the client's city boutique; guardrail on ‘日内瓦的秋天’ patterns.

Policy applied to the wrong product · case A05, option 3

v1 Unacceptable

唐先生，不客气。手镯若有需要调整的地方，三十天内可以免费改圈；往后清洁抛光也不收费用。您方便的时候提前告诉我，我安排一个安静时段，不必等。

手镯“三十天内免费改圈”——政策只适用于戒指

v2 Unacceptable

唐先生，手镯戴了一周，若尺寸上想再合手一些，30天内可以免费改圈。您来湖滨店前跟我说一声，我替您安排个安静些的时段。

手镯“30天免费改圈”——与v1相同的错误

v3 OK

唐先生，手镯若有需要，清洁抛光随时可以做，我这边帮您约个方便的时间，到店或我送过去都行。您忙的话，不回复也没关系。

Change: Fact sheet states ‘ring only’; guardrail checks resizing against rings.

Invented store exhibition · case RA06, option 3

v2 Unacceptable

冯女士，小珍宝系列的四叶造型其实来自瓦洛蕾1920年代的档案图稿，去年给您女儿选的那枚珍珠母贝也是这个来源。最近店里有些关于老图稿的小展陈，您如果对这类工艺故事感兴趣，可以来看看，不一定要买东西，路过就好。

编造“店里有些关于老图稿的小展陈”

v3 OK

冯女士，小珍宝那枚四叶的造型来自1920年代的档案图稿，忽然想起您女儿那枚。她这一年过得可好？您忙您的，不用特意回我，改天想清洁了跟我说一声就是。

Change: Fact sheet + critic; guardrail on exhibitions not present in the input.

Ignoring ‘no event invitations’ · case E03, option 3

v2 Unacceptable

杨女士，好久没联系。11月28日周六下午店里有一场Constellation高级珠宝私享预览，我想到您对当代艺术的眼光，觉得您或许会有兴趣——但也知道您一向偏爱安静的私人时段。
所以三个选择：那天来，我全程陪着；另约一个只有您的时间；或者我先把这场的资料发给您看。您回一个数字就好。

明确邀请“那天来，我全程陪着”，违反客户偏好

v3 OK

杨女士，想到您对当代艺术的眼光，Constellation高级珠宝私享预览或许合您心意。11月28日周六15:00–18:00在静安恒汇店，不过按您的习惯，我更想为您留一段安静的一对一私享时段。
若那阵子不便，我也可以先把作品的图片或视频发给您，看过再约。您看哪种方式轻松些？

Change: CRM preference translated into a hard instruction: offer only a one-to-one appointment.

E3. What remains to be fine-tuned

Remaining issue	Evidence	Next step
Critic edits can leave a sentence without its point	A06-1, RA02-3	Critic returns a fluent rewrite and the result is checked again; if the fix would remove the angle's core, regenerate that option instead of deleting phrases.
Critic removes legitimate boutique names	R01, RA08 (“湖滨店”, “静安”)	Put the boutique and CA facts in the fact sheet.
Date arithmetic	RA03-2 “四年多” (5 years 8 months)	Pre-compute durations for every purchase and conversation in the fact sheet.
Small invented pretexts	R08-1, RA05-3	A whitelist of honest openers (“想到您…”) and a soft rule; measure with the pretext detector.
Old thread referenced ambiguously	A08-3	Pass only the gist of closed threads, not their wording.
Personalisation is the weakest dimension	judge 3.93 / 5	Richer CRM memory (past chats, occasions, family dates) through retrieval; few-shot examples from real top-performing CAs; ultimately fine-tune on approved messages.
Evaluation bias	Same model family judges; test set written by the tool's author	Second judge family (Qwen), a panel of real CAs for inter-rater agreement, cases sampled from anonymised CA logs, a larger red-team set.
Single sample per case	Live re-runs of the same case produce different drafts	Evaluate 3–5 samples per case and report the worst case and variance, not one draw; consider a lower temperature for sensitive scenarios.
No real-world signal yet	—	Online metrics: which option the CA picks, edit distance before sending, client reply rate, opt-outs.
Latency and cost	6.7 s mean, 2 calls	Stream the first option; run the critic in parallel per option; cache the brand and system prompt.
Timing only warns	A08	Schedule the send for the next allowed window.

Bonus. Architecture

See the diagram in B2 (source: `docs/architecture.svg`). Code map: `context.py` (step 1) · `prompts.py`, `scenarios.py` (2) · `pipeline.py` (2–4, repair) · `guardrails.py` (4) · `judge.py`, `scripts/` (offline loop). Earlier versions are frozen in `snapshots/`.